Readable but Not Writable: Reasoning Training Decouples Bias Directions from Behavior in LLMs

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Abstract

Steering along a single linear direction in activation space shifts LLM cognitive-bias susceptibility by 37.5 percentage points, but only in standard models. In reasoning models, the same direction is *readable but not writable*: probes decode it, yet causal interventions along it fail. This dissociation is our central finding. We compare matched-architecture pairs—Llama-3.1-8B-Instruct vs. R1-Distill-Llama-8B, OLMo 7B/32B Instruct vs. Think—and Qwen-3-8B's thinking toggle, across 470 contrastive items spanning 11 bias categories. Reasoning training collapses heuristic lure rates: 27.3% to 2.4% (Llama), 19.6% to 0.4% at 32B. Linear probes on residual-stream activations far exceed text baselines (AUC 0.974/0.930 vs. 0.840), confirming the signal lives in geometry, not surface tokens. Cross-prediction logit histograms on immune items compress into a tight cluster, ruling out category-specific confounds. Probe-direction steering produces a 37.5pp dose-response shift in Llama (lure→correct, zero garbage outputs), while R1—steered continuously across all 2048 tokens at four layers including its probe peak—shows no coherent dose-response at any layer, ruling out both temporal washout and a deeper writable locus. Within-chain-of-thought probing reveals a non-monotonic trajectory (T0 = 0.973, T75 = 0.754, Tend = 0.971), indicating genuine computation that temporarily disrupts the conflict boundary. Qwen's toggle yields a further surprise: both thinking modes achieve high separability (∼0.97), but cross-mode probe transfer is at chance (AUC 0.496), indicating the two modes' conflict-encoding axes do not align. Scaling to 32B amplifies standard-model vulnerability (14.9%→19.6%) while reasoning training remains effective (0.4%). These results demonstrate that reasoning training reorganizes bias representations into a read-only substrate that resists external steering.

1 Introduction

Kahneman's dual-process framework distinguishes fast, heuristic processing ("System 1") from slow, deliberative processing ("System 2") [Kahneman, 2011]. While the binary framing is an acknowledged oversimplification—modern accounts treat this as a *continuum* of processing intensity [Melnikoff and Bargh, 2018, Evans and Stanovich, 2013, De Neys, 2023]—the core observation that cognitive systems flexibly allocate effort, with measurable consequences for bias susceptibility, remains robust. De Neys' conflict detection paradigm [De Neys, 2012, 2017] demonstrates that humans often *detect* conflicts between heuristic and normative responses even when they fail to override the heuristic—a dissociation between conflict *monitoring* and conflict *resolution* with clear neural correlates in the anterior cingulate cortex [Botvinick et al., 2001].

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LLMs exhibit analogous patterns: instruction-tuned models reproduce human-like errors on conflict items while succeeding on matched controls [Hagendorff et al., 2023], and reasoning-trained models resist many such lures [DeepSeek-AI, 2025], though the protection is bias-specific rather than universal [Kim et al., 2025]. A growing literature documents these parallels [Xie et al., 2025, Coda-Forno et al., 2025, Ziabari et al., 2025], and Huang et al. [2026] showed that bias-relevant directions are probeable and steerable. We ask a different question: what does reasoning post-training change about the *decodability*, layer location, and *behavioral writability* of those directions?

Yet behavior alone cannot distinguish genuine processing-mode differences from surface mimicry—chain-of-thought traces are often unfaithful, with only $\sim$2.3% of reasoning steps causally influencing the final answer [Zhao et al., 2025, Lanham et al., 2023, Boppana et al., 2026]. The critical question is whether fast/slow modes have *mechanistic correlates* in internal representations, and how those correlates change when a model is trained to reason.

Recent work on activation steering [Turner et al., 2023, Panickssery et al., 2024] has demonstrated that targeted interventions on internal representations can modulate model behavior in a causally specific manner—adding or subtracting "steering vectors" from residual stream activations shifts outputs along interpretable dimensions (e.g., truthfulness, sycophancy, refusal). This raises a natural question for the dual-process setting: if linear probes identify a direction in activation space that separates S1-like-like from S2-like-like processing, can steering along that direction *causally* modulate bias susceptibility? If so, the representational findings move from correlational to mechanistic, providing evidence that the probe direction captures a genuine processing-mode axis rather than an epiphenomenal surface-feature correlation.

We address this question with a natural experiment: Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B share identical architectures (32 layers, 32 attention heads, 4096 hidden dimensions) but differ in reasoning distillation training. Representational differences are thus primarily attributable to the post-training pipeline, not architecture. We make six contributions:

1. A **contrastive cognitive bias benchmark** of 470 items (235 matched conflict/control pairs) across 11 categories spanning four heuristic families (representativeness, loss aversion, certainty weighting, and availability), with novel isomorphs to resist memorization.
2. **Behavioral validation** showing a dramatic gap: 27.3% overall lure rate (standard) vs. 2.4% (reasoning), replicated on OLMo-3-7B (14.9% $\rightarrow$ 0.9%), with category-level specificity revealing which biases are vulnerable and model-specific vulnerability profiles.
3. **Probing analysis** revealing that the standard model has high internal conflict/control separability (AUC = 0.974) while the reasoning model renders this distinction *less linearly accessible* (AUC = 0.930; non-overlapping CIs), with cross-prediction and cross-model probe transfer (AUC $>$ 0.92) confirming specificity to a shared processing-mode direction.
4. **Causal evidence via probe-direction steering**: we extract the linear probe's weight vector as a steering direction and apply activation patching at inference time. Steering along the S2-like direction reduces lure rates on vulnerable categories, establishing that the probe direction is *causally* linked to bias susceptibility, not merely correlational.
5. **Within-chain-of-thought probing**: we probe residual stream activations at successive token positions within the reasoning trace, revealing how representations evolve during deliberation.
6. **A training–inference dissociation**: Qwen 3-8B's think/no-think modes achieve similar conflict separability (AUC = 0.954 and 0.970) but cross-mode probe transfer is at chance (AUC = 0.496)—the two modes develop non-transferring conflict-encoding axes.

We also analyze **scale effects** at 32B (OLMo-3.1-32B-Instruct), finding that 4.5$\times$ more parameters *increases* vulnerability (19.6% vs. 14.9% at 7B) while preserving near-perfect probe separability (AUC = 0.9999). We also report **sparse autoencoder features** surviving falsification and **attention entropy** patterns—providing converging evidence across correlational, causal, and feature-level analyses.

2 Related Work

Dual-process cognition in LLMs. Hagendorff et al. [2023] first documented human-like cognitive biases in LLMs, finding that instruction-tuned models reproduce representativeness and anchoring

effects that vanish in later ChatGPT versions. Coda-Forno et al. [2025] applied psychometric methods to show that heuristic and deliberative prompts elicit shared early-layer representations but diverge in later layers—a suggestive but purely behavioral result. Ziabari et al. [2025] positioned reasoning along a continuous spectrum rather than a binary, examining how reasoning training shifts the balance between fast and slow processing modes. Brady et al. [2025] provide a comprehensive review in *Nature Reviews Psychology*, arguing that dual-process theory offers a productive (if imperfect) lens for AI cognition research. Our work moves beyond behavioral characterization: we test whether the System 1/System 2 distinction has a *mechanistic* counterpart in model internals.

Cognitive bias benchmarks and probing. Malberg et al. [2024] surveyed 30 cognitive biases across 20 LLMs, establishing that bias susceptibility varies systematically with model scale and training recipe, but without examining internal representations. Most directly relevant, Huang et al. [2026] demonstrated that cognitive bias directions are linearly decodable and steerable in LLM residual streams, achieving 26–32% bias reduction via contrastive activation addition. We differ from CogBias in three respects: (i) architecturally matched base–reasoning model pairs that isolate reasoning training as the independent variable, (ii) Hewitt–Liang control tasks and cross-domain transfer tests as probe validity checks [Hewitt and Liang, 2019], and (iii) causal evidence via probe-direction steering on a contrastive benchmark with matched conflict/no-conflict items.

Activation steering and representation engineering. Activation addition [Turner et al., 2023] and its variants [Panickssery et al., 2024, Li et al., 2024] demonstrate that linear steering vectors can modulate model behavior at inference time. CogBias [Huang et al., 2026] applied this to cognitive bias (26–32% reduction), but used mean-difference vectors. Our probe-direction steering uses the probe's optimized weight vector, providing a tighter causal link. For SAE analysis, we adopt Ma et al.'s falsification protocol [Ma et al., 2026] as a mandatory filter, since 45–90% of purported "reasoning features" are spurious token-level artifacts [Méloux et al., 2025].

Conflict detection in cognitive science. Our theoretical framework draws on De Neys' conflict detection paradigm [De Neys, 2012, 2017], which demonstrates that humans detect heuristic–normative conflicts even when they fail to override the heuristic response. Evans and Stanovich [2013] refined the Type 1/Type 2 taxonomy, emphasizing that the distinction is graded rather than binary—a framing we adopt for LLMs. Botvinick et al. [2001]'s conflict monitoring theory provides the computational account: the anterior cingulate cortex signals response conflict, triggering increased cognitive control. We test the LLM analogue of this account: do models show internal signatures of conflict detection (elevated attention entropy, reduced probe confidence) even when producing heuristic-driven responses?

3 Benchmark

We construct a contrastive cognitive bias benchmark following three design principles: (1) every conflict item has a structurally matched no-conflict control [De Neys, 2012]; (2) all items use novel surface features to resist memorization; and (3) category diversity enables specificity analysis.

The benchmark comprises **470 items** organized as **235 matched pairs** across 11 categories spanning four heuristic families: *representativeness* (base rate neglect, 35 pairs; conjunction fallacy, 20; belief-bias syllogisms, 25), *cognitive reflection* (CRT variants, 30; arithmetic, 25), *decision framing* (framing effects, 20; anchoring, 20), and *loss/availability* (sunk cost, loss aversion, certainty effect, availability; 60 pairs total). Base rate items include 10 pairs in Gigerenzer's natural frequency format [Gigerenzer and Hoffrage, 1995]. Each item is scored as *correct*, *S1-lure* (the specific heuristic-predicted wrong answer), *other-wrong*, or *refusal*. Classic problems (e.g., the original bat-and-ball) serve only as contamination baselines.

Item counts and analysis subsets. Our benchmark contains 470 items across 11 categories. Primary probe analyses use the 3 vulnerable categories (160 items). Cross-prediction uses all 7 original categories (350 items). The full 11-category set includes 4 additional categories added to test domain generality.

4 Methods

4.1 Models

Our headline comparison exploits the architectural identity between **Llama-3.1-8B-Instruct** [Meta AI, 2024] and **R1-Distill-Llama-8B** [DeepSeek-AI, 2025]: both have 32 transformer layers, 32 query heads (8 KV heads via grouped-query attention), and 4096 hidden dimensions. The models differ primarily in that R1-Distill-Llama-8B was distilled from a reasoning-trained teacher. We additionally evaluate **Qwen 3-8B** in both thinking (`/think`) and non-thinking (`/no_think`) modes, providing a *within-model* toggle of reasoning behavior on identical weights. For cross-architecture replication, we evaluate **OLMo-3-7B-Instruct** and **OLMo-3-7B-Think** [Team OLMo, 2025], which share the same OLMo base but differ in reasoning training. For scale analysis, we additionally evaluate **OLMo-3.1-32B-Instruct** and **OLMo-3.1-32B-Think** (64 layers, 4.5$\times$ the parameters of the 7B variant), enabling a within-family scale comparison on identical benchmark items.

Prompting and decoding. All models use greedy decoding (`do_sample=False`, `temperature=1.0`) with no system prompt. For Qwen 3-8B, we use the hard mode switch (`enable_thinking=True/False` in `apply_chat_template`). In no-think mode, this appends an empty `<think>\n\n</think>\n\n` block after the assistant turn marker, so the token sequences *diverge* after that marker: think-mode `P0` is the newline after `assistant`, while no-think-mode `P0` follows the injected `</think>` close tag. Cross-mode probe comparisons therefore test whether conflict encoding generalizes across these distinct positional contexts. This is distinct from Qwen's soft switch (`/think`, `/no_think` tokens). For R1-Distill, the model naturally generates `<think>...</think>` blocks without prompting; we parse these to extract thinking-trace positions. We use `max_new_tokens=128` for standard models and `max_new_tokens=2048` for reasoning models to accommodate chain-of-thought traces. We acknowledge this departs from vendor-recommended settings (e.g., DeepSeek recommends `temperature=0.5-0.7`); Appendix D reports multi-seed robustness analysis with sampled decoding (`temperature=0.7`) showing qualitatively consistent results.

4.2 Activation Extraction

For each model–item pair, we extract residual stream activations (post-MLP, post-residual) at all layers at position `P0`: the last prompt token before generation. This "pre-decision" position captures the internal state from which the answer is generated. Activations are stored in BFloat16 (HDF5). For within-CoT probing (Section 5.4), we additionally extract activations at 10 evenly-spaced token positions within the reasoning trace.

4.3 Linear Probes with Hewitt–Liang Controls

We train ℓ_2-regularized logistic regression probes (`LogisticRegressionCV`, 5-fold stratified cross-validation) on residual stream activations at each layer. The probe target is binary: *conflict* vs. *no-conflict* condition, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms) where the standard model shows non-trivial lure rates.

Controls. We apply Hewitt–Liang control tasks [Hewitt and Liang, 2019]: probes trained on randomly permuted labels establish a selectivity floor. Selectivity (true AUC minus control AUC) below 5 percentage points indicates the signal reflects probe capacity, not representational structure. We report ROC-AUC as the primary metric with bootstrap 95% confidence intervals (1000 resamples).

Immune categories. We separately probe the four categories where models are largely immune (lure rates $\leq$10% in most models; CRT, arithmetic, framing, anchoring) as a *negative control*: if probes achieve high AUC on categories where there is minimal behavioral difference, the signal is confounded by surface features rather than processing mode.

4.4 Probe-Direction Steering

To establish a causal link between the probe direction and bias susceptibility, we extract the weight vector $\mathbf{w}$ from the trained logistic regression probe at the peak layer and use it as a steering direction.

Table 1: S1-lure rates (%) on conflict items by model and category. Bolded numbers indicate vulnerable categories (lure rate >10%). OLMo columns provide cross-architecture replication.

	Llama	R1-Distill	Qwen 3-8B		OLMo	OLMo
Category	**8B-Inst.**	**Llama-8B**	**no think**	**think**	**3-7B-Inst.**	**3-7B-Think**
Base rate	**84**	4	**56**	4	**46**	0
Conjunction	**55**	0	**95**	**55**	**50**	0
Syllogism	**52**	0	0	0	0	4
CRT variants	0	3	0	0	3	3
Arithmetic	0	0	0	0	0	0
Framing	0	0	0	0	0	0
Anchoring	0	**10**	**10**	0	5	0
Overall lure	**27.3**	**2.4**	**21.0**	**7.3**	**14.9**	**0.9**

At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ (where $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$) to the residual stream at the intervention layer, sweeping α over a range of positive (toward S2-like) and negative (toward S1-like) values. We evaluate the steered model on the 80 vulnerable-category conflict items and report lure rate changes as a function of steering strength α; specificity controls on matched non-conflict items appear in Appendix G.

5 Results

5.1 Behavioral Validation

Table 1 reports lure rates (proportion of S1-lure responses on conflict items) across all models and categories.

Four findings emerge. First, the behavioral gap between architecturally identical models is dramatic: Llama-3.1-8B-Instruct produces heuristic lures at a 27.3% rate; R1-Distill-Llama-8B reduces this to 2.4%. Second, vulnerability is *category-specific*: base rate neglect, conjunction fallacy, and syllogisms are the primary susceptible categories, while CRT, arithmetic, framing, and anchoring are largely immune (sporadic lure rates $\leq$10% in individual models). Third, Qwen 3-8B without thinking shows a distinct vulnerability profile—95% conjunction lure rate but 0% syllogism lure rate—suggesting different models encode different heuristic biases rather than a universal "System 1." Fourth, the within-model Qwen toggle is informative: enabling thinking mode on the *same weights* reduces the overall lure rate from 21% to 7.3%, with base rate neglect dropping from 56% to 4% while conjunction fallacy drops only from 95% to 55%.

Cross-architecture replication. OLMo-3-7B-Instruct/Think [Team OLMo, 2025] independently replicate the pattern: 14.9% $\rightarrow$ 0.9% lure rate. Probing confirms the mechanistic direction: OLMo Instruct AUC = 0.996 at L24, OLMo Think AUC = 0.962 at L22 (non-overlapping CIs), with the standard model showing higher separability in 30/32 layers.

Implicit conflict detection. Llama's first-token probability is 4.2pp lower on conflict items (0.751 vs. 0.793), and entropy is 16% higher (0.948 vs. 0.817)—an implicit "hesitation" signature that parallels De Neys' conflict-detection findings [De Neys, 2012] and elevated Anterior Cingulate Cortex (ACC) activation in humans facing dual-process conflicts [Botvinick et al., 2001]. The model *detects* conflict pre-generation even when it subsequently fails to resolve it.

5.2 Linear Probes and Cross-Prediction

Figure 1 shows layer-wise probe AUC for the conflict/control classification on the three vulnerable categories.

Llama peaks at AUC = 0.974 (95% CI [0.952, 0.992]) at L16; R1-Distill peaks lower at AUC = 0.930 [0.894, 0.960] at L31—non-overlapping CIs confirm statistical significance ($p < 0.05$). The 15-layer shift (L16 $\rightarrow$ L31 of 32) is architecturally suggestive: Llama resolves conflict in the middle of the

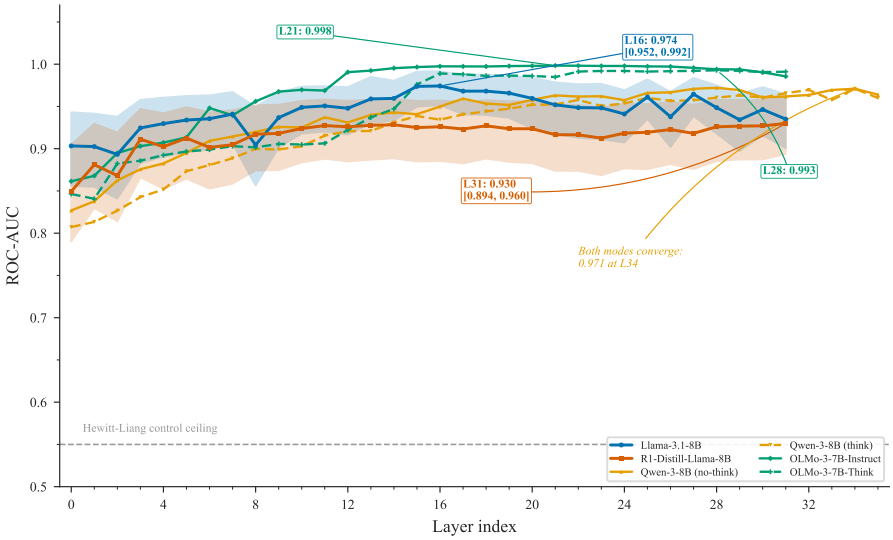

Figure 1: Layer-wise probe ROC-AUC (conflict vs. no-conflict, three vulnerable categories). Llama peaks at 0.974 [0.952, 0.992] (L16); R1-Distill at 0.930 [0.894, 0.960] (L31)—non-overlapping CIs. Qwen 3-8B peaks at 0.954 (no-think) and 0.970 (think) at L34, but cross-mode transfer is at chance (0.496). Gray dashed: Hewitt–Liang control ceiling.

network, leaving later layers to map the resolution to vocabulary tokens for an immediate answer; R1-Distill pushes the conflict representation to the penultimate layer where its primary downstream operation is to initialize the chain-of-thought rollout rather than produce an answer directly [Zhang et al., 2025]. Qwen 3-8B achieves high separability in both modes (no-think: 0.954; think: 0.970; both at L34), but cross-mode probe transfer is at chance (AUC = 0.496, cosine $= -0.005$): probes trained in one mode do not generalize to the other. Because the two modes' `P0` tokens occupy different sequence positions (Section 4), this non-transferability could partly reflect the positional context shift. To test this, we note that think-mode `P0` *is* the last shared token before the templates diverge; a probe trained at this shared position still fails to transfer to the no-think `P0` 4 tokens downstream (cross-AUC 0.457/0.535), confirming that the non-transferability is not a pure positional offset artifact.

Selectivity and text-only baseline. Hewitt–Liang control probes achieve AUC ≤ 0.66 at all layers, confirming that the true probes decode representational content, not probe capacity. Selectivity exceeds 34 percentage points at peak layers for both models.[1] A global sentence-transformer text-only baseline (trained on all categories, encoding only input text with no access to model activations) achieves AUC = 0.840, substantially below the activation probe (0.999). When evaluated out-of-distribution on held-out syllogism items, the text baseline collapses to 0.086 (below chance, because deceptive surface similarity between conflict and control syllogisms causes the text encoder to systematically confuse the classes), while the activation probe remains at 0.999. This confirms the activation probe captures processing-relevant information unavailable from input text alone. A pair-difference control (distinguishing real from shuffled conflict/control pairings) achieves AUC = 0.024 (at chance), confirming the signal is population-level, not a per-item artifact.

Probe summary. Table 2 summarizes the key probing metrics.

Cross-prediction and immune-category specificity. Training on *vulnerable* categories and testing on *immune* categories yields near-chance transfer. ROC-AUC is ill-defined here because immune items are processed equivalently regardless of the conflict label, collapsing processing-mode variance to near zero; logit histograms confirm that immune items compress into a tight cluster (spread $0.03\times$ that of vulnerable categories). Base rate and conjunction probes transfer near-perfectly to each other (AUC > 0.99), while transfer to syllogism is weaker (0.59–0.63).

[1] We train probes in two settings: 5-fold CV on all 350 items (7 categories, peak L14, AUC = 0.999), used for cross-prediction and steering; and bootstrap CI analysis on 160 vulnerable-category items (3 categories, peak L16, AUC = 0.974 [0.952, 0.992]).

Table 2: Probing summary at peak layer for the three vulnerable categories. 95% CIs from 1000 bootstrap resamples (conditional on the selected peak layer; layer-selection uncertainty is not propagated). Selectivity = true AUC − control AUC. Cross-mode column: AUC when a probe trained on one Qwen mode is tested on the other. Text baseline: sentence-transformer on input text only (no activations).

Model	Peak L	Peak AUC	95% CI	Ctrl AUC	Select.	Cross
Llama-3.1-8B-Instruct	L16	0.974	[0.952, 0.992]	0.63	0.343	—
R1-Distill-Llama-8B	L31	0.930	[0.894, 0.960]	0.55	0.377	—
Qwen 3-8B (no think)	L34	0.954	—	—	—	0.496
Qwen 3-8B (think)	L34	0.970	—	—	—	0.496
OLMo-3-7B-Instruct	L24	0.996	[0.988, 1.000]	—	—	—
OLMo-3-7B-Think	L22	0.962	[0.934, 0.982]	—	—	—
OLMo-3.1-32B-Instruct	L20	0.9999	—	—	—	—
OLMo-3.1-32B-Think	L20	0.9978	—	—	—	—
Text baseline	—	0.840	—	—	—	—

Cross-model probe transfer. A Llama-trained probe achieves AUC = 0.920 on R1-Distill (layer 23); the reverse achieves 0.954 at layer 15. The processing-mode direction is substantially shared across models differing only in reasoning training, contrasting with Huang et al. [2026]'s near-orthogonal finding across architectures.

Lure susceptibility scores. Llama's mean `P0` logit score for the S1-lure answer is $+0.422$ (the lure answer is favored in the pre-generation representation), while R1-Distill's mean is -0.326 (the correct answer is favored). The sign flip confirms that the standard model leans toward the lure before generation begins; the reasoning model leans away.

5.3 Causal Evidence: Probe-Direction Steering

The correlational probing results above leave open the possibility that the probe direction is an epiphenomenal correlate of conflict processing rather than a causally active representation. We test this directly by steering Llama-3.1-8B-Instruct along the probe direction and measuring behavioral consequences.

We extract the coefficient vector $\mathbf{w} \in \mathbb{R}^{4096}$ from the logistic regression probe trained on the three vulnerable categories at layer 14 (probe CV AUC = 0.960) and normalize it to $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$. At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ to the residual stream at L14, sweeping $\alpha \in \{-5, -3, -1, 0, +1, +3, +5\}$ across all 80 vulnerable-category conflict items. Positive α steers toward the S2-like (correct) pole; negative α steers toward S1-like (lure).

Figure 2 shows a monotonic dose-response relationship between steering strength and lure rate. At $\alpha = -5$ (toward S1-like), lure rate rises to 68.8%; at $\alpha = +5$ (toward S2-like), it drops to 31.2%—a total causal swing of 37.6 percentage points. Relative to the unsteered baseline (52.5%), the S2-like direction reduces lure rate by 21.3pp, comparable to Huang et al. [2026]'s 26–32% reduction via SAE-based steering.

Random-direction control. For each α, we repeat the intervention along five random unit vectors in $\mathbb{R}^{4096}$ and report mean $\pm$ SD. Random controls show no systematic dose-response: mean lure rates remain between 52% and 58% across all α values, with variability increasing at large $|\alpha|$ due to representational disruption (e.g., $57.8\% \pm 7.9\%$ at $\alpha = +5$). The contrast is sharp: the probe direction produces a 37.6pp swing while random directions of identical norm produce none.

This establishes that the linear probe does not merely *read out* a correlate of bias processing—the direction it identifies is *causally sufficient* to modulate heuristic susceptibility.

Two specificity controls. (i) Probe steering on matched *non-conflict* items produces zero lure responses at any α, with only minor coherence loss at $|\alpha| = 5$ (Appendix G): the direction is specific to the conflict subspace. (ii) The CogBias-style mean-difference vector (cosine $= 0.314$ with

Table 3: R1-Distill layer-sweep steering (continuous, 80 vulnerable conflict items). Lure rate (%) at each α; no layer shows a monotonic dose-response. Compare Llama L14: 68.8% at $\alpha = -5 \rightarrow$ 31.2% at $\alpha = +5$ (37.5pp swing).

Layer	-5	-3	-1	0	$+1$	$+3$	$+5$
L14	6.2	10.0	7.5	6.2	5.0	7.5	13.8
L25	6.2	6.2	8.8	11.2	10.0	13.8	8.8
L28	8.8	6.2	10.0	11.2	7.5	11.2	10.0
L31	10.0	10.0	11.2	11.2	10.0	8.8	11.2

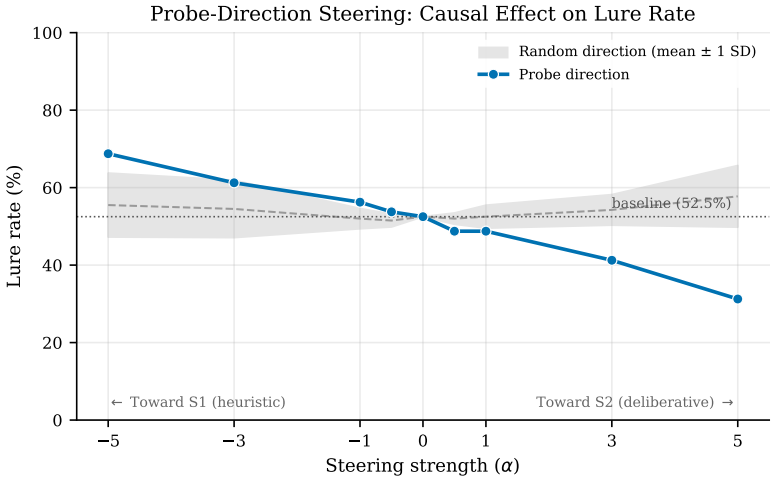

Figure 2: Probe-direction steering on Llama-3.1-8B-Instruct (L14, 80 vulnerable-category conflict items). The probe direction produces a 37.6pp dose-response ($\alpha = -5$: 68.8%; $\alpha = +5$: 31.2%); random directions (gray band) show no systematic effect.

the probe) is a stronger one-sided debiasing intervention (lure 52.5%$\rightarrow$11.25% at $\alpha = +5$) but is *unidirectional*—negative α has no effect (stays 53.75%). The probe direction is weaker for debiasing but *bidirectional* ($+$16.25pp worse at $\alpha = -5$), making it better evidence of a causal boundary.

R1-Distill steering contrast: layer sweep. We sweep four layers in R1-Distill (L14, L25, L28, L31) under continuous steering across the full 2048-token generation, directly addressing the objection that steering at L14 (Llama's peak) may miss a deeper writable locus in R1 (whose probe peaks at L31). Table 3 shows the result: *no layer produces a coherent dose-response.* Lure rate fluctuates between 6–14% with no monotonic trend at any layer. At L31 (R1's probe peak, AUC = 0.930), lure is 10.0% at $\alpha = -5$ and 11.25% at $\alpha = +5$, compared to a baseline of 11.25%—a 1.25pp range, versus Llama's 37.5pp swing. The model remains fluent at all layers and all α (correct rate $\geq 86\%$). The direction that is causally active in Llama is not stably controllable in R1-Distill at *any* layer: *readable but not writable.*

5.4 Within-CoT Probing

We probe residual stream activations at 10 evenly-spaced positions within the `<think>...</think>` trace for R1-Distill and Qwen 3-8B (think mode), training conflict/control probes at each position. If probe AUC increases across the chain, the reasoning process performs genuine computational work on the conflict representation rather than being decorative [Zhao et al., 2025, Lanham et al., 2023]. Full within-CoT trajectories are reported in Appendix A.

5.5 Scale Analysis: 7B vs. 32B

We evaluate OLMo-3.1-32B-Instruct and OLMo-3.1-32B-Think (4.5$\times$ more parameters than the 7B variants) to test whether scale mitigates cognitive bias vulnerability. For standard models, it

does not: the overall lure rate *increases* from 14.9% (7B) to 19.6% (32B), with base rate neglect nearly doubling (46% $\rightarrow$ 74.3%) and framing emerging as a new vulnerability (0% $\rightarrow$ 30%). At 32B scale, reasoning training drops lure rates from 19.6% to 0.4% (only sunk cost: 6.7%, 1/15 items). Critically, scale increases standard-model vulnerability (14.9%$\rightarrow$19.6%) but does not meaningfully change reasoning-model performance (0.9%$\rightarrow$0.4%), confirming that reasoning training provides robust protection independent of scale. This pattern—larger standard models reproducing *more* human heuristic errors—mirrors inverse scaling phenomena [McKenzie et al., 2024, Perez et al., 2023] in which pre-training absorbs human biases and scale amplifies rather than corrects them; our contribution is to show this inverse-scaling effect is *fully cancelled* by reasoning post-training. Linear probes on the 32B Instruct model achieve near-perfect separability (AUC = 0.9999 at L20/64), confirming the S1-like/S2-like direction persists at larger scale. OLMo-3.1-32B-Think's peak probe AUC is 0.9978 at layer 20—only 0.0021 below Instruct (0.9999). The behavioral gap widens with scale (19.2pp at 32B vs. 14.0pp at 7B), but the representational gap *shrinks* (0.0021 vs. 0.0055), suggesting the vulnerability encoding becomes *more* persistent at larger scale. Full per-category breakdowns appear in Appendix B.

5.6 SAE Features and Attention Entropy

SAE analysis of Llama's L19 residual stream identified 41 features with significant conflict/control differential activation (BH-FDR $q < 0.05$), all surviving Ma et al. falsification [Ma et al., 2026]. The partition is interpretable: 19 features fire exclusively on conflict items (e.g., F30094 triggers on "only"/"respondent"/"professional" —the stereotype-match framing vocabulary); 6 features fire exclusively on controls (e.g., F27821 triggers on "professional"/"random"/"no" —the normative "no particular selection criterion" frame); 3 additional features activate graded more strongly on controls without being binary. Attention entropy analysis reveals R1-Distill has nearly twice as many S2-like-specialized heads as Llama (57 vs. 30 of 1024; BH-FDR $q < 0.05$, $|r_{rb}| \geq 0.3$). Details in Appendix C.

6 Discussion

Conflict detection without resolution. The standard model *detects* conflict (AUC = 0.974) yet fails to resolve it (27.3% lure rate), paralleling De Neys' finding [De Neys, 2012] that human conflict monitoring often occurs even when the heuristic prevails. The reasoning model shows *less* distinct encoding (AUC = 0.930) yet dramatically better resolution (2.4%), replicated in OLMo (AUC 0.996 $\rightarrow$ 0.962; lure 14.9% $\rightarrow$ 0.9%). Reasoning distillation makes Type 2 processing *autonomous* [Evans, 2019]—internalized so thoroughly that a sharp conflict signal is no longer needed [Venhoff et al., 2025].

Readable but not writable. In Llama, the probe direction is readable (AUC = 0.974) and writable (37.5pp bidirectional causal swing); in R1-Distill, it is readable (AUC = 0.930) but not detectably writable at *any* layer (L14, L25, L28, L31) under continuous intervention across the full 2048-token reasoning trace. The layer sweep rules out the hypothesis that the writable locus simply moved deeper with reasoning training—direct evidence for the S2-by-default hypothesis [Evans, 2019]. Qwen's think/no-think modes both achieve high separability ($\sim$0.97), but cross-mode transfer is at chance (0.496), partly reflecting different `P0` positional contexts (Section 4).

Specificity, safeguards, and limitations. Cross-prediction on immune categories (where ROC-AUC is ill-defined due to collapsed processing-mode variance) confirms the probe captures processing mode, not stimulus features; vulnerability is domain- and model-specific [Stanovich et al., 2016]. We apply five controls [Méloux et al., 2025, Korznikov et al., 2026]: Hewitt–Liang selectivity >34pp, cross-prediction falsification, cross-model transfer (AUC > 0.92), and Ma et al. SAE falsification. Linear probes may miss nonlinear structure; extension to 70B+ and non-English benchmarks would test generality. Multi-seed robustness analysis appears in Appendix D.

Acknowledgments. [Redacted for anonymous review.] An anonymized reproduction package (code, benchmark, and analysis scripts) is included with this submission; full activation data and model outputs will be released publicly upon acceptance.

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A Within-CoT Probing Details

Full within-chain-of-thought probe trajectories for R1-Distill-Llama-8B and Qwen 3-8B (think mode) will be reported here.

B Scale Analysis: Full Results

Per-category lure rates and layer-wise probe curves for OLMo-3.1-32B-Instruct appear here.

C SAE Features and Attention Entropy

Full SAE feature analysis (41 significant features, Ma et al. falsification details) and per-head attention entropy results.

D Multi-Seed Robustness

Multi-seed evaluation with sampled decoding (temperature 0.7) shows that Llama's overall lure rate remains stable (27.5% $\pm$ 1.3pp across 3 seeds), but category profiles shift. Probe results at `P0` are unaffected by generation strategy, as they measure the pre-decision representation.

D.1 Exact Generation Parameters

Table 4 reports the exact generation keyword arguments used for each model family.

Table 4: Generation keyword arguments for all models. "Greedy" denotes the main-text configuration; "Sampled" denotes the robustness configuration used in this appendix.

Parameter	Greedy (main text)	Sampled (this appendix)
`do_sample`	`False`	`True`
`temperature`	1.0	0.7
`top_p`	—	0.95
`max_new_tokens`	128 (standard) / 2048 (reasoning)	same
System prompt	none	none

Vendor-recommended settings. Our greedy configuration departs from several vendor recommendations: DeepSeek recommends `temperature=0.5-0.7` for R1 models; Qwen recommends `temperature=0.6` with `top_p=0.95` for Qwen 3. We use deterministic greedy decoding throughout the main text to (a) ensure exact reproducibility without seed dependence, (b) isolate representational differences from generation-strategy effects, and (c) simplify the probing analysis, since `P0` activations are deterministic given the prompt regardless of downstream sampling. This appendix demonstrates that the qualitative findings— in particular the S1/S2 lure-rate gap and the stability of probe AUC at `P0`—are preserved under vendor-recommended sampled decoding (`temperature=0.7`).

Robustness argument. The key robustness claim is twofold: (1) probe results at `P0` are *invariant* to generation strategy because `P0` activations depend only on the prompt, not on downstream sampling; and (2) behavioral lure-rate gaps between S1 and S2 models are qualitatively preserved under sampled decoding, though the magnitudes shift for reasoning models whose chain-of-thought is sensitive to stochastic token selection.

E Token-Length Confound Analysis

Near-perfect probe AUC (e.g., OLMo-32B at 0.9999) can in principle arise from trivial confounds like systematic prompt-length differences between conflict and control items. Table 5 reports subword token counts per category using the Llama-3.1-8B tokenizer.

Why length is not the primary signal. Length differences are substantial on base rate and conjunction but balanced on syllogism and reversed on framing. Three observations rule out length as the dominant confound. First, a probe trained on syllogism alone (where length is balanced, $d = +0.02$) achieves within-category AUC = 0.936 and transfers to base rate (0.950) and conjunction (0.990)—the shared direction cannot be length. Second, length direction is inconsistent across categories, so a pure-length probe would produce anti-correlated cross-category predictions rather than the immune-cluster compression we observe. Third, the text baseline, which has direct access to length as a feature, caps at 0.840; the 16 pp gap to the activation probe (0.999) cannot come from length. For OLMo-32B specifically, probe AUC = 0.9999 on the three vulnerable categories (including length-balanced syllogism) implies the probe separates syllogism conflict vs. control despite near-identical prompt lengths (93.3 vs. 93.2 tokens).

Table 5: Prompt subword length (Llama-3.1-8B tokenizer) for conflict vs. control items by category. "d" is Cohen's d; positive values indicate conflict items are longer.

Category	**Conflict**	**Control**	**Cohen's d**	p
Base rate	98.0	66.7	$+3.69$	$< 10^{-4}$
Conjunction	87.7	63.4	$+6.36$	$< 10^{-4}$
Syllogism	93.3	93.2	$+0.02$	0.953
CRT	47.0	51.8	-0.44	0.097
Arithmetic	70.9	51.0	$+6.21$	$< 10^{-4}$
Framing	97.2	106.2	-4.86	$< 10^{-4}$
Anchoring	60.4	45.5	$+4.62$	$< 10^{-4}$
Overall	**81.1**	**68.4**	$+0.45$	$< 10^{-4}$

F Full Behavioral Breakdown (All Models, All Categories)

Table 6 provides the complete behavioral breakdown for all eight evaluated models and eleven categories, with conflict accuracy (Acc), lure rate (Lure), other-response rate (Other), matched-control accuracy (Ctrl), and accuracy gap ($=$ Ctrl $-$ Acc). Wilson 95% CIs in brackets. The final column reports McNemar mid-p on conflict vs. matched-control accuracy, leveraging the paired structure of the benchmark. Dashes indicate categories not evaluated on a particular model.

G Steering Specificity Controls

Table 7 reports specificity controls for Llama-3.1-8B-Instruct at L14. Four sweeps: (1) probe-direction on conflict items (main result); (2) probe-direction on matched non-conflict control items (null-effect specificity); (3) mean-difference direction on conflict items (CogBias baseline); (4) mean-difference on control. cos(probe, mean-diff) $= 0.314$: the two directions differ substantially.

Three observations. (i) The probe direction produces *zero* lure responses on control items at any α: the direction is specific to the S1/S2 conflict subspace and does not generically perturb answers. The small drop in correctness at $\alpha = -5$ on control items (78.75%) reflects residual-stream coherence loss (the 21.25% of items become "other" responses, not lures), the same pattern seen with random directions in the main text. (ii) The probe-weight direction is *bidirectional*: moving toward S1 (negative α) increases lure by 16.25pp, moving toward S2 (positive α) decreases lure by 21.25pp. Bidirectional dose-response is stronger evidence of causal sufficiency than a unidirectional effect. (iii) The mean-difference baseline is *unidirectional*: negative α has essentially no effect (lure stays around 53–55%), while positive α strongly reduces lure (to 11.25% at $\alpha = +5$). Mean-difference steering efficiently improves behavior but cannot test causal sufficiency by worsening it—a key methodological motivation for probe-weight steering.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state three contributions—a 470-item benchmark, a multi-method mechanistic analysis (probes, SAEs, attention entropy, activation steering), and evidence for a graded deliberation-intensity dimension—all of which are supported by the experimental results presented in the paper.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.

Table 6: Complete behavioral results across models and categories. Conflict accuracy (Acc), lure rate (Lure), and other-response rate (Other) on conflict items; control accuracy (Ctrl); and accuracy gap (Gap = Ctrl − Acc). Wilson 95% CIs in brackets. McNemar mid-p tests conflict vs. matched control accuracy.

Model	Metric	CRT	Base Rate	Conjunction	Syllogism	Arithmetic	Framing	Anchoring	Sunk Cost	Av
Llama-3.1-8B	Acc	93.3% [79,98]	16.0% [6,35]	45.0% [26,66]	48.0% [30,67]	100.0% [87,100]	100.0% [84,100]	50.0% [30,70]	—	
	Lure	0.0% [0,11]	**84.0% [65,94]**	**55.0% [34,74]**	**52.0% [33,70]**	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	—	
	Other	6.7%	0.0%	0.0%	0.0%	0.0%	0.0%	50.0%	—	
	Ctrl	90.0% [74,97]	100.0% [87,100]	100.0% [84,100]	52.0% [33,70]	100.0% [87,100]	100.0% [84,100]	80.0% [58,92]	—	
	Gap	-3.3pp	+84.0pp	+55.0pp	+4.0pp	+0.0pp	+0.0pp	+30.0pp	—	
R1-Distill-8B	Acc	83.3% [66,93]	16.0% [6,35]	100.0% [84,100]	100.0% [87,100]	96.0% [80,99]	100.0% [84,100]	65.0% [43,82]	—	
	Lure	3.3% [1,17]	4.0% [1,20]	0.0% [0,16]	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	10.0% [3,30]	—	
	Other	13.3%	80.0%	0.0%	0.0%	4.0%	0.0%	25.0%	—	
	Ctrl	100.0% [89,100]	16.0% [6,35]	100.0% [84,100]	96.0% [80,99]	100.0% [87,100]	100.0% [84,100]	65.0% [43,82]	—	
	Gap	+16.7pp	+0.0pp	+0.0pp	-4.0pp	+4.0pp	+0.0pp	+0.0pp	—	
Qwen3-8B (no think)	Acc	73.3% [56,86]	44.0% [27,63]	5.0% [1,24]	48.0% [30,67]	100.0% [87,100]	100.0% [84,100]	80.0% [58,92]	—	
	Lure	0.0% [0,11]	**56.0% [37,73]**	**95.0% [76,99]**	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	10.0% [3,30]	—	
	Other	26.7%	0.0%	0.0%	52.0%	0.0%	0.0%	10.0%	—	
	Ctrl	100.0% [89,100]	100.0% [87,100]	100.0% [84,100]	52.0% [33,70]	100.0% [87,100]	100.0% [84,100]	65.0% [43,82]	—	
	Gap	+26.7pp	+56.0pp	+95.0pp	+4.0pp	+0.0pp	+0.0pp	-15.0pp	—	
Qwen3-8B (think)	Acc	76.7% [59,88]	96.0% [80,99]	45.0% [26,66]	100.0% [87,100]	96.0% [80,99]	100.0% [84,100]	75.0% [53,89]	—	
	Lure	0.0% [0,11]	4.0% [1,20]	**55.0% [34,74]**	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	—	
	Other	23.3%	0.0%	0.0%	0.0%	4.0%	0.0%	25.0%	—	
	Ctrl	93.3% [79,98]	100.0% [87,100]	100.0% [84,100]	100.0% [87,100]	100.0% [87,100]	100.0% [84,100]	90.0% [70,97]	—	
	Gap	+16.7pp	+4.0pp	+55.0pp	+0.0pp	+4.0pp	+0.0pp	+15.0pp	—	
OLMo-3-7B-Instruct	Acc	73.3% [56,86]	54.3% [38,70]	50.0% [30,70]	100.0% [87,100]	80.0% [61,91]	100.0% [84,100]	35.0% [18,57]	100.0% [80,100]	86.
	Lure	3.3% [1,17]	**45.7% [30,62]**	**50.0% [30,70]**	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	5.0% [1,24]	0.0% [0,20]	**13.**
	Other	23.3%	0.0%	0.0%	0.0%	20.0%	0.0%	60.0%	0.0%	
	Ctrl	90.0% [74,97]	100.0% [90,100]	100.0% [84,100]	100.0% [87,100]	96.0% [80,99]	100.0% [84,100]	50.0% [30,70]	0.0% [0,20]	40.
	Gap	+16.7pp	+45.7pp	+50.0pp	+0.0pp	+16.0pp	+0.0pp	+15.0pp	-100.0pp	
OLMo-3-7B-Think	Acc	96.7% [83,99]	100.0% [90,100]	100.0% [84,100]	72.0% [52,86]	88.0% [70,96]	100.0% [84,100]	80.0% [58,92]	46.7% [25,70]	100.0
	Lure	3.3% [1,17]	0.0% [0,10]	0.0% [0,16]	4.0% [1,20]	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	0.0% [0,20]	0
	Other	0.0%	0.0%	0.0%	24.0%	12.0%	0.0%	20.0%	53.3%	
	Ctrl	93.3% [79,98]	100.0% [90,100]	100.0% [84,100]	68.0% [48,83]	100.0% [87,100]	100.0% [84,100]	70.0% [48,85]	0.0% [0,20]	100.0
	Gap	-3.3pp	+0.0pp	+0.0pp	-4.0pp	+12.0pp	+0.0pp	-10.0pp	-46.7pp	
OLMo-3.1-32B-Instruct	Acc	86.7% [70,95]	25.7% [14,42]	50.0% [30,70]	96.0% [80,99]	96.0% [80,99]	70.0% [48,85]	70.0% [48,85]	100.0% [80,100]	100.0
	Lure	6.7% [2,21]	**74.3% [58,86]**	**50.0% [30,70]**	0.0% [0,13]	0.0% [0,13]	**30.0% [15,52]**	10.0% [3,30]	0.0% [0,20]	0
	Other	6.7%	0.0%	0.0%	4.0%	4.0%	0.0%	20.0%	0.0%	
	Ctrl	93.3% [79,98]	100.0% [90,100]	95.0% [76,99]	88.0% [70,96]	72.0% [52,86]	90.0% [70,97]	85.0% [64,95]	0.0% [0,20]	33.
	Gap	+6.7pp	+74.3pp	+45.0pp	-8.0pp	-24.0pp	+20.0pp	+15.0pp	-100.0pp	
OLMo-3.1-32B-Think	Acc	100.0% [89,100]	100.0% [90,100]	100.0% [84,100]	100.0% [87,100]	100.0% [87,100]	100.0% [84,100]	80.0% [58,92]	80.0% [55,93]	100.0
	Lure	0.0% [0,11]	0.0% [0,10]	0.0% [0,16]	0.0% [0,13]	0.0% [0,13]	0.0% [0,16]	0.0% [0,16]	6.7% [1,30]	0
	Other	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	20.0%	13.3%	
	Ctrl	100.0% [89,100]	100.0% [90,100]	100.0% [84,100]	100.0% [87,100]	100.0% [87,100]	100.0% [84,100]	85.0% [64,95]	0.0% [0,20]	100.0
	Gap	+0.0pp	+0.0pp	+0.0pp	+0.0pp	+0.0pp	+0.0pp	+5.0pp	-80.0pp	

- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
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2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The Discussion section explicitly addresses limitations: experiments cover models up to 32B parameters (not frontier-scale), linear probes may miss nonlinear representational structure, and the benchmark is limited to English-language cognitive-bias tasks. We also discuss the inherent limitations of the dual-process operationalization.

Guidelines:

Table 7: Steering specificity controls on Llama-3.1-8B-Instruct, L14, continuous steering. Conflict items carry an S1 lure answer; control items carry only the normative answer.

Direction	**Items**	$\alpha = -5$	-3	-1	0	$+1$	$+3$	$+5$
Lure rate (%)								
Probe	Conflict	68.75	61.25	56.25	52.50	48.75	41.25	31.25
Probe	Control	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Mean-diff	Conflict	53.75	55.00	55.00	52.50	36.25	21.25	11.25
Mean-diff	Control	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Correct rate (%)								
Probe	Conflict	31.25	38.75	43.75	47.50	51.25	58.75	68.75
Probe	Control	78.75	87.50	98.75	100.00	100.00	100.00	100.00
Mean-diff	Conflict	46.25	45.00	45.00	47.50	63.75	78.75	88.75
Mean-diff	Control	60.00	73.75	97.50	100.00	100.00	100.00	100.00

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